

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The distance formula

1. A circle has its centre at $(2a, a - 7)$ and passes through the point $(11, -9)$. Its diameter is $10\sqrt{2}$ units. Find the values of a . **CBSE 2025, 2 marks**

SOLUTION

Radius = $\frac{10\sqrt{2}}{2} = 5\sqrt{2}$ units, so the distance from the centre to $(11, -9)$ is $5\sqrt{2}$.

$$(2a - 11)^2 + (a - 7 + 9)^2 = (5\sqrt{2})^2$$

$$4a^2 - 44a + 121 + a^2 + 4a + 4 = 50, \text{ so } 5a^2 - 40a + 75 = 0$$

$$a^2 - 8a + 15 = 0, \text{ so } (a - 3)(a - 5) = 0$$

Answer: $a = 3$ or $a = 5$

2. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD . (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units **CBSE 2024, 1 mark**

SOLUTION

D is the mid-point of BC : $D = \left(\frac{6+0}{2}, \frac{4+0}{2}\right) = (3, 2)$

$$AD = \sqrt{(5-3)^2 + (-6-2)^2} = \sqrt{4+64} = \sqrt{68}$$

Answer: (a)

3. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x . (a) 12, -18 (b) -12, 18 (c) 18, 5 (d) -9, -12 **CBSE 2024, 1 mark**

SOLUTION

Both points have the same y -coordinate, so the distance is $|x - 3| = 15$.

$$x - 3 = 15 \text{ or } x - 3 = -15, \text{ so } x = 18 \text{ or } x = -12.$$

Answer: (b)

2 The section formula

4. Where does the mid-point of the segment joining $P(-4, 5)$ and $Q(4, 6)$ lie? **CBSE 2025, 1 mark**

SOLUTION

$$\text{Mid-point} = \left(\frac{-4 + 4}{2}, \frac{5 + 6}{2} \right) = \left(0, \frac{11}{2} \right)$$

Its x -coordinate is 0.

Answer: on the y -axis

5. In what ratio does the y -axis divide the segment joining $(5, -6)$ and $(-1, -4)$? Find the point where they meet. **CBSE 2025, 3 marks**

SOLUTION

Let the ratio be $k : 1$. The point of division is $\left(\frac{-k + 5}{k + 1}, \frac{-4k - 6}{k + 1} \right)$.

On the y -axis the x -coordinate is 0: $-k + 5 = 0$, so $k = 5$.

$$y = \frac{-4(5) - 6}{5 + 1} = -\frac{26}{6} = -\frac{13}{3}$$

Answer: ratio 5 : 1; the point is $\left(0, -\frac{13}{3} \right)$

6. A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ **CBSE 2024, 1 mark**

SOLUTION

The centre is the mid-point of the diameter. Let the other end be (x, y) .

$$\frac{6 + x}{2} = 2 \text{ gives } x = -2, \text{ and } \frac{0 + y}{2} = 0 \text{ gives } y = 0.$$

Answer: (c)

7. The point $\left(\frac{8}{5}, y \right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment? Find y as well. **CBSE 2024, 3 marks**

SOLUTION

Let the ratio be $k : 1$. Then $\frac{2k + 1}{k + 1} = \frac{8}{5}$.

$10k + 5 = 8k + 8$, so $k = \frac{3}{2}$. The ratio is 3 : 2.

$$y = \frac{3 \times 3 + 2 \times 2}{3 + 2} = \frac{13}{5}$$

Answer: ratio 3 : 2 and $y = \frac{13}{5}$

8. $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P , Q , R and S are the mid-points of AB , BC , CD and DA . Show that the diagonals of $PQRS$ bisect each other.

CBSE 2024, 3 marks

SOLUTION

$$P = \left(-1, \frac{5}{2}\right), Q = (1, 6), R = \left(3, \frac{5}{2}\right), S = (1, -1)$$

$$\text{Mid-point of diagonal } PR = \left(\frac{-1+3}{2}, \frac{5/2+5/2}{2}\right) = \left(1, \frac{5}{2}\right)$$

$$\text{Mid-point of diagonal } QS = \left(\frac{1+1}{2}, \frac{6-1}{2}\right) = \left(1, \frac{5}{2}\right)$$

The diagonals PR and QS have the same mid-point, so they bisect each other.

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